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organic matter into basic gases such as synthetic gas—a mix of hydrogen and carbon monoxide gas. This synthetic gas is almost a green fuel, which is used by advanced gas turbines for the generation of electrical power.

After purification process, SynGas is often cleaner than the natural gas (mixture of hydrocarbon gases). Its heating value depends on the type of feed treated, but typically SynGas has a lower heating value of 10-15 megajoules per kilogram.

2. The inorganic materials are simultaneously melted into molten slag, which upon cooling becomes a vitrified, inert glass-like material (through magnavication process) that can be used by the construction industry. The homogenous and sulphuric products contained in the feed are transformed respectively to HCl (hydrochloric acid), HF (hydrofluoric acid) and H₂S (hydrogen sulphide). Suitable neutralisation techniques for these three products are employed. No ashes are produced in the process.

An artistic view of complete processing of trash (feed) and generation of byproducts using the plasma gasification technology is shown in Fig. 4.

Components of plasma gasification system

The design of plasma gasification plant is not standardised and various companies engaged in their manufacture custom build the facility as per customers' specifications. However, following are some of the essential parts of any such facility:

Trituration and conveyor section. This section comprises a grinder or crusher for breaking the trash into manageable size for handling by the conveyor, which pushes it to the plasma furnace.

Pyrolysis chamber with plasma torch/torches. It is an air-locked chamber with one or more plasma torches. It allows the garbage in but prevents hot gases from escaping into the atmosphere. The chamber is lined

with heat-resistant refractory material. Towards the bottom side of the chamber is drainage system for slag. It also incorporates a water-cooling system. Towards the top of chamber is the exit vent for the gases.

Gas reformer and heat exchanger. The reformer reforms hydrocarbon fuel into a reformat gas such as hydrogen-rich syn gas. In the heat exchanger, the hot gases heat water to produce steam, which is then used for running a steam turbine coupled to a generator to produce electricity.

Gas cleanup filter. As the gases produced by pyrolysis mainly comprise carbon monoxide (25 per cent), hydrogen and hydrocarbons (15 per cent), and carbon dioxide and nitrogen (60 per cent), the next step is to 'clean' the syngas or producer gas. Thereafter, the gas is burned in internal combustion (IC) engine generator sets or turbines to produce electricity.

Pyrolysis and gasification: similarities and differences

Pyrolysis is thermal degradation of waste in the absence of air to produce char, pyrolysis oil and syngas, e.g. the conversion of wood into charcoal.

Gasification is breakdown of hydrocarbons into a syngas by carefully controlling the amount of oxygen present.

Both pyrolysis and gasification turn wastes into energy-rich fuels by heating the waste under controlled conditions. Whereas incineration fully converts the input waste into energy and ash, these processes deliberately limit the conversion so that combustion does not take place directly. Instead, they convert the waste into valuable intermediates that can be further processed for materials recycling or energy recovery.

Caution. It is rather early to draw conclusions about toxicity effects of the emissions from both these processes, because the true impact of most chemicals and the mixtures of chemicals are still poorly understood.

To be concluded next month